

GUJARAT TECHNOLOGICAL UNIVERSITY

BE- SEMESTER-I & II EXAMINATION – WINTER 2024

Subject Code:BE01000061

Date:07-01-2025

Subject Name:Engineering Graphics & Design

Time:10:30 AM TO 01:00 PM

Total Marks:70

Instructions:

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) Give illustration and application of type of lines	03
	(b) A line PQ 70 mm long has its end P in VP and end Q in HP. Line is inclined to HP by 60° and VP by 30° . Draw the projections	04
	(c) Construct a diagonal scale of R.F. = $1/36$ showing yard, foot and inch. Scale should be long enough to measure 5 yards. Indicate on it 3 yards 2 feet 9 inches	07
Q.2	(a) A 30° - 60° set square has its shortest side 50 mm long and is in the H.P. The Top view of the set square is isosceles triangle. Draw the projections of the set square find its inclination with the H.P.	03
	(b) Construct a scale 1:50 to read up to 6 metres in meters and decimetres. Show on it a distance of 4.9 metres.	04
	(c) The major and minor axes of an ellipse measure 100 mm and 70 mm respectively. Draw an ellipse by the rectangle method	07
OR		
	(c) A line AB has a point P on it such that $AP : PB = 1 : 2$. The end A is in the first quadrant and it is 20 mm above H.P. while the end B is in the V.P. The point P is 35 mm from the H.P. The line is inclined at 30° to the H.P. and the elevation length of the line is 70 mm. Draw the projections of the line AB and the point P. Find the true length, the plan length and the inclination of the line with V.P.	07
Q.3	(a) Draw following projection of points: (1) Point 'A' is 20 mm above HP and 25 mm In front of VP. (2) Point 'B' is 35 mm behind VP and 30 mm above HP. (3) Point 'C' is in the HP and 15 mm In front of VP.	03
	(b) Give Conic Terminology with sketch	04
	(c) An isosceles triangle ABC having its base $AB = 40$ mm and altitude 60 mm is resting on the H.P. on Its base AB. Draw the projections of the plane when its surface is inclined to H.P. at an angle of 45° and the base AB which is on the H.P. is making an angle of 50° to the V.P	07
OR		
Q.3	(a) Give classification of Engineering Curves	03
	(b) A cylinder, diameter of base 60 mm and height 70 mm, is having a point of its periphery of base in V.P. with axis of cylinder inclined to V.P. by 45° and parallel to H.P. or ground. Draw the projections of the cylinder.	04

- (c) A cone, diameter of base 60 mm and height 70 mm, has one of its generators in H.P. and making an angle of 45° with the V.P. Draw the projections of the cone when the apex is towards the observer. **07**
- Q.4** (a) Draw the following sketches:
Truncated Cylinder, Frustum of a cone and Frustum of a square pyramid. **03**
- (b) A rectangular plate PQRS. 25 mm x 40 mm size, is in space with shorter edge parallel to H.P. and 15 mm above it. Plate PQRS is perpendicular to V.P. and inclined to H.P. by such an angle so that its plan becomes square. Draw the projections and find out its inclination with H.P. **04**
- (c) A pentagonal pyramid, side of base 40 mm and height 80 mm, is resting on H.P. on its base with one of the edges of the base away from V.P. is parallel to VP. It is cut by an A.I.P. bisecting the axis, the distance of the section plane from the apex being 20 mm. Draw the elevation and sectional plan of the pyramid and draw the true shape of the section. Find the inclination of the section plane with the H.P. **07**
- OR**
- Q.4** (a) Give different types of section planes **03**
- (b) ABCD is a rhombus of diagonals AC 110 mm and BD 70 mm. Its corner A is in the H.P. and the plane is inclined to the H.P. such that the plan appears to be a square. Draw the projections of the plane and find its inclination with H.P. **04**
- (c) Two lemons on a tree, planted near the compound wall of a bungalow, are 1.0 m and 1.25 m above the ground and 0.5 m and 0.75 m from a 15 cm thick compound wall but on the opposite sides of it. The distance between lemons measured along the ground and parallel to the wall is 1.0 m. Determine the real distance between centre of two lemons. **07**
- Q.5** (a) A semi-circular thin plate, of 60 mm diameter, rests on the H. P. on its diameter, the surface is inclined at 30° to the H. P. Draw the projections of the plate. **04**
- (b) Fig-1 shows pictorial view of an object with using particular using any one dimensioning method. Draw following views by first angle projection method **10**
(a) Sectional elevation at AA (b) Top view (c) Right hand side view
- OR**
- Q.5** (a) (1) Give Symbols for Methods of Projection
(2) Give difference between to two dimensional method with drawing. **04**
- (b) Fig-2 shows pictorial view of an object with using particular any one dimensioning method. Draw following views by first angle projection method **10**
(a) Sectional elevation at AA (b) Top view (c) Left hand side view

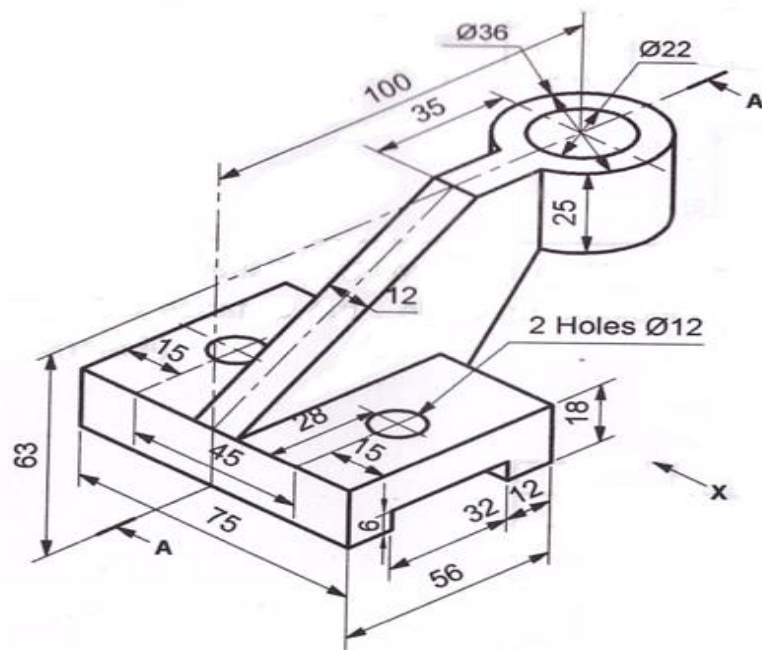


Fig.-1

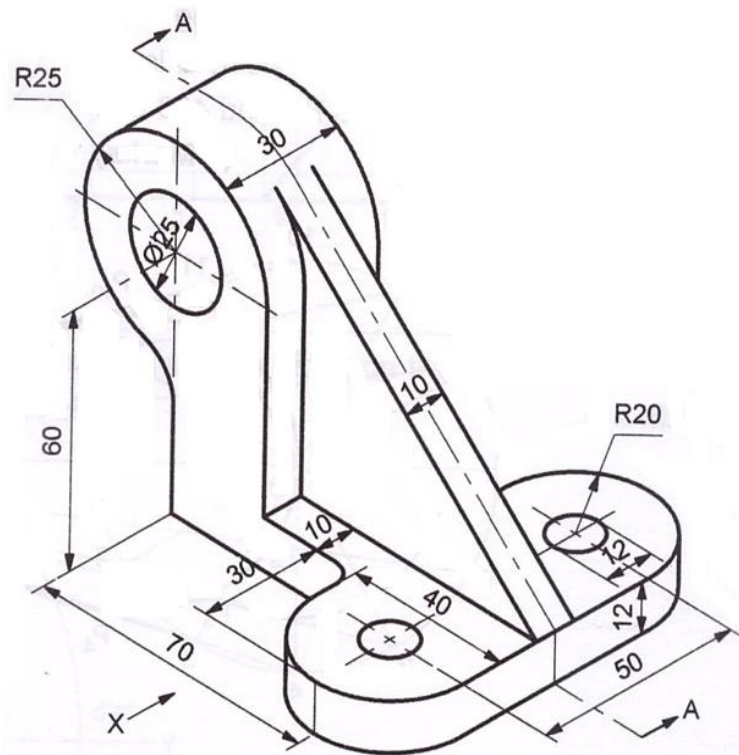


Fig-2

GUJARAT TECHNOLOGICAL UNIVERSITY**BE- SEMESTER-I&II EXAMINATION – SUMMER 2025****Subject Code:BE01000061****Date:10-06-2025****Subject Name:Engineering Graphics & Design****Time:10:30 AM TO 01:00 PM****Total Marks:70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		Marks
Q.1	(a) Draw regular pentagon and hexagon of 30 mm side using UNIVERSAL METHOD	03
	(b) The distance between two towns is 250 km and is represented by a line of length 50mm on a map. Construct a scale to read 600 km and indicate a distance of 530 km on it.	04
	(c) Draw an Archimedian spiral of 1.5 convolutions, the largest & the smallest radius being 55 mm & 10 mm respectively. Draw tangent & normal to the spiral at a point 40 mm from the pole	07
Q.2	(a) Draw following projection of points: (1) Point 'A' is 25 mm above HP and 20 mm Infront of VP. (2) Point 'B' is 35 mm behind VP and 40 mm above HP. (3) Point 'C' is in the HP and 10 mm Infront of VP	03
	(b) Draw projections of line AB=85 mm inclined to 60° HP & 30° VP	04
	(c) The distance between the end projectors of a straight-line PQ is 130mm. The end P is 40mm below HP and 25mm in front of the VP. Q is 75 mm above HP and 30 mm behind VP. Draw its projections. Find TL of the line	07
	OR	
	(c) The elevation of line AB, 80 mm long, measures 55 mm. The end A is 20 mm above H.P. and 10 mm in front of V.P. Draw the projection of the line and finds its true inclination with H.P. and V.P, if the end B is 25 mm below H.P. and is behind V.P	07
Q.3	(a) Enlist types of lines.	03
	(b) ABC is a triangle of sides AB = 75 mm, BC = 60 mm and CA = 45 mm. Its longest side is in H.P. Its surface makes an angle of 45°with the H.P. Draw its projections.	04
	(c) A pentagonal plate, side 25 mm is resting on H.P. on one of its corners with opposite edge to the corner making 30° with V.P. The plate is inclined to H.P. by 45°. Draw the projections of regular pentagonal plate	07
	OR	
Q.3	(a) Define the following curves : involute, ellipse and cycloid.	03
	(b) A hexagonal plane of 30 mm side, rests on the V.P. on an edge such that the surface is inclined at 45° to the V.P. Draw its projections	04
	(c) A circular plane having the diameter 50 mm is resting with point A of its periphery on H.P. The surface of the plane is inclined to H.P. such that the plan of the plane becomes an ellipse with minor axis 30mm.Draw the projection of the plane when the plan of the diameter through point A is inclined at 30o to V.P.and the centre of the plane is 40mm from V.P.Find the inclination of the plane with H.P.	07

- Q.4 (a)** Classify solids with shapes of drawings **03**
(b) Differentiate between aligned system and unidirectional system of dimensioning **04**
(c) A square pyramid, side of base 35 mm and axis length 50 mm is lying on the H.P. on one of its triangular faces. Draw the projections of the pyramid when the base edge contained by the triangular face on H.P. makes an angle of 45° on the V.P. keeping apex of the pyramid towards the observer **07**

OR

- Q.4 (a)** Differentiate between **03**
 1) Square pyramid and tetrahedron
 2) Cube and square prism
(b) A cone 40 mm diameter and 50 mm axis is resting on one generator on HP. Draw its projections **04**
(c) A hexagonal prism, edge of base 20 mm and axis 50 mm long, rests with its base on HP such that one of its rectangular faces is parallel to VP. It is cut by a plane perpendicular to VP, inclined at 45° to HP and passing through the right corner of the top face of the prism. (i) Draw the sectional top view. **07**
- Q.5 (a)** Define Representative Fraction. With reference to Representative Fraction, classify the scales **03**
(b) Differentiate between prism and pyramid **04**
(c) Fig. 1 shows pictorial view of an object. Draw Sectional Elevation from – X and Plan. In the figure 1, the arrow shows “X” direction. **07**

OR

- Q.5 (a)** Enlist methods for drawing Ellipse. **03**
(b) Write the comparison between first angle projection method and third angle projection method **04**
(c) Draw the front view looking from X –direction and plan of a object shown in Figure 2 using first angle projection method **07**

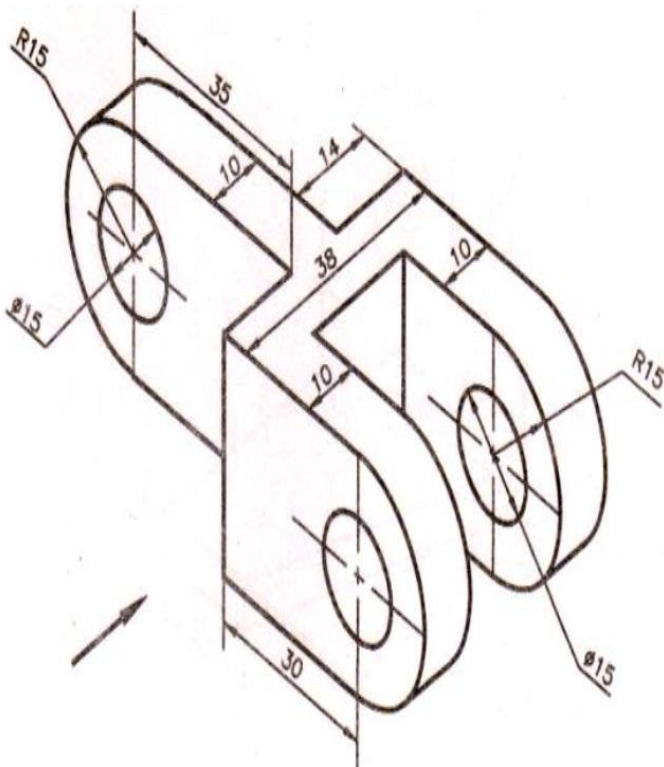


Figure 1

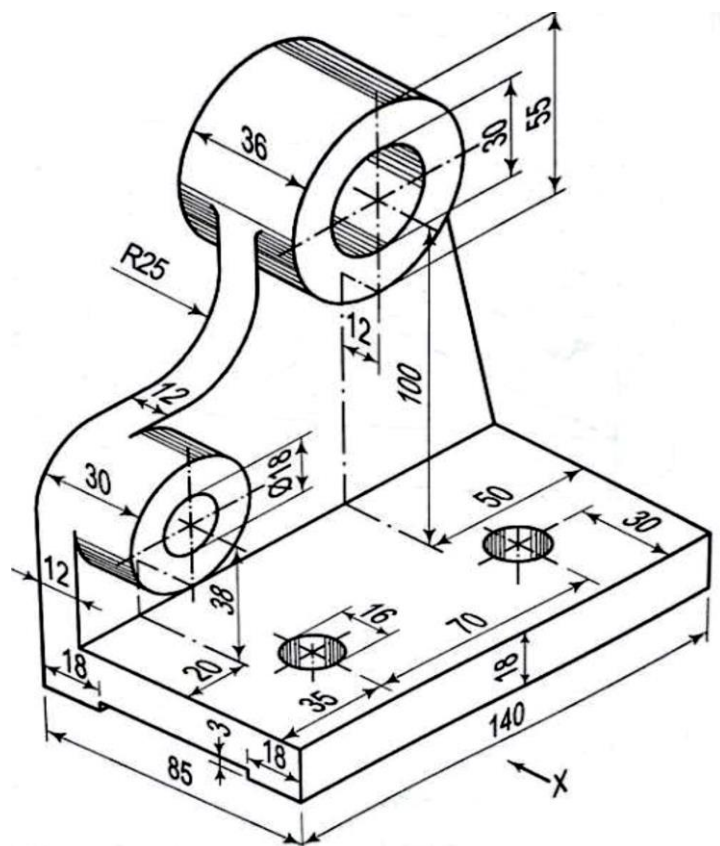


Figure 2

Enrollment No./Seat No.:

GUJARAT TECHNOLOGICAL UNIVERSITY

Bachelor of Engineering - SEMESTER - 1/2 EXAMINATION - WINTER 2025

Subject Code: BE01000061/BE01R00061

Date: 26-12-2025

Subject Name: Engineering Graphics & Design

Time: 02:30 PM TO 05:00 PM

Total Marks: 70

Instructions

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

	Marks
Q.1 (a) Divide 100 mm long line into 7 equal parts.	03
(b) Classify engineering curves. Give at least one application of each conic curve.	04
(c) (1) Define Representative Fraction and State use of enlarged scales. (2) Draw diagonal scale of R.F. 1:2.5, showing centimeters and millimeters and long enough to measure up to 20 cm. Show the distance 13.4 cm on it.	07
Q.2 (a) Draw the projections of the following points on the same xy line keeping the projectors 25 mm apart. (1) Point A is in HP and 20 mm behind VP. (2) Point B is 40 mm above HP and 25 mm in front of VP. (3) Point C is in VP and 40 mm above HP.	03
(b) Draw the projections of a 100 mm long line PQ which is inclined at an angle of 45° to H.P. and 30° to V.P. One of its end point 'P' is in H.P. as well as in VP.	04
(c) Draw an ellipse by concentric circle method given the major and minor axis are 120 mm and 80 mm respectively.	07
OR	
(c) Construct logarithmic spiral for one convolution. Given the length of the shortest radius vector equal to 10 mm and the ratio of the lengths of the successive radius vectors equal to 7/5 for vectorial angle of 45°.	07
Q.3 (a) The distance between the end projectors of line AB is 60 mm. Line AB has its end A 20 mm above H.P. and 25 mm in front of V.P. The other end B is 45 mm above H.P. and 55 mm in front of V.P. Draw projections of line AB.	03
(b) Draw the following 2D planes with given dimensions. (1) Circle with radius of 2.5 cm. (2) Regular pentagon with side X = 30 mm. (3) Regular hexagon with side 30 mm.	04
(c) ABCD is a rhombus of diagonals AC = 110 mm and BD = 70 mm. Its corner A is in the H.P. and the plane is inclined to H.P. such that the plan appears to be a square. The plan of diagonal AC makes an angle of 20° to the V.P. Draw the projections of the plane and find its inclination with H.P. (Hint. Take a scale of 1:2)	07

OR

- (a) (1) Define 2nd quadrant. 03
(2) Define Auxiliary Inclined Plane (AIP).
(3) Draw F.V. and T.V. of 30 mm long line AB perpendicular to H.P. and parallel to V.P. Take point A at any distance from H.P. & V.P.
- (b) A circular plate of diameter 50 mm is resting on H.P. on a point on the circumference with its surface inclined at 45° to H.P. and perpendicular to V.P. Draw its projections. 04
- (c) A square plate PQRS of side 35 mm is resting on corner P with diagonal PR making 30° with H.P. and diagonal QS inclined to V.P. by 60° and parallel to H.P. Draw the projections of the square plate PQRS. 07

- Q.4 (a) Classify solids in detail. 03
- (b) (1) Explain concept of orthographic projection. 04
(2) Classify orthographic projections.
- (c) A hexagonal pyramid, side of base 25 mm and axis 55 mm long, has one of its slant edges on the H.P. This slant edge is inclined at 45° to the V.P. Draw its projections when the apex is nearer to V.P. 07

OR

- (a) Name the solid. 03
(1) I am a pyramid with triangular base having all sides of base and faces are equal.
(2) I am a prism with square base having all sides of bases and faces are equal.
(3) I have two equal size circular bases.
- (b) Write four differences between first angle and third angle projection methods. 04
- (c) A cone, base diameter 80 mm and axis length 80 mm is resting on its base on the H.P. It is cut by an AIP inclined at 45° to the H.P. and cutting the axis at 30 mm from the apex. Draw front view, sectional top view and true shape of the section. 07

- Q.5 (a) Write three industrial applications of AutoCAD. 03
- (b) Using first angle projection method, draw F.V. and T.V. of the object given in Figure 1. 04

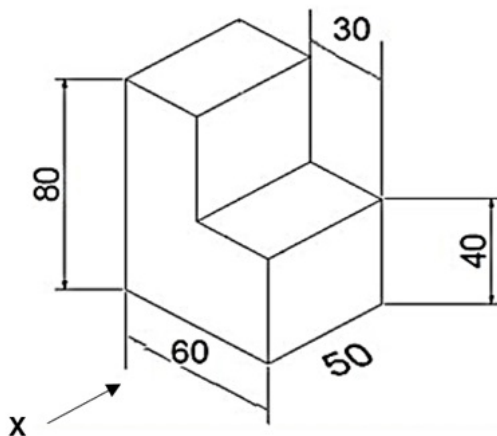


Figure 1 (Que. 5 (b))

- (c) List the seven essential commands of modify panel in AutoCAD, state their usage and explain their execution in workspace. 07

OR

- (a) Write three advantages of AutoCAD over conventional manual drawing. 03
- (b) Using third angle projection method, draw F.V. and R.H.S.V. of the object given in Figure 2. 04

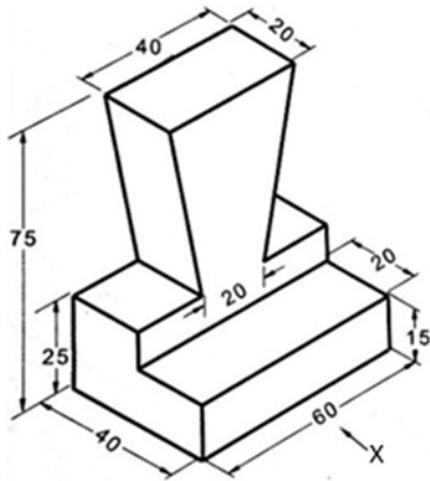


Figure 2 (Que. 5(b) OR

- (c) Write name of the following AutoCAD command symbols, state their use and explain their execution. 07

- (1) 
- (2) 
- (3) 
- (4) 
- (5) 
- (6) 
- (7) 
